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Identifying conserved trajectories in clonal evolution trees

Background

The analysis of clonal evolution provides an option to reconstruct tumor development over time, explore treatment response, and identify potential reasons for therapy failure. This allows for an in-depth analysis of a patient's heterogeneous tumor cell composition. Furthermore, phylogenetic trees hold the promise of revealing common patterns conserved across different patients.

To identify conserved trajectories in clonal evolution trees, several approaches exist, e.g. TreeMHN [1] or MASTRO [2]. However, TreeMHN requires mutations to be acquired successively. While it is usually possible to obtain this information from single-cell DNA-sequencing data, it is often impossible to determine the exact order of two mutations in bulk sequencing data. Natural variation in the coverage and thus, insecurities in the calculated variant allele frequencies (VAFs) and cancer cell fractions (CCFs) hamper the deconvolution process. This commonly results in several mutations being assigned to the same clone.

The tool MASTRO does support mutations of unknown order. However, the resulting high number of possible relations hinders the analysis of more complex phylogenetic trees due to substantial storage requirements. Additionally, to our knowledge, all currently available tools performing an analysis on conserved evolutionary trajectories focus exclusively on mutated genes, neglecting copy number variants (CNVs). This is expected to cause a substantial bias to the results.

Methods

We propose a novel approach for identifying conserved trajectories in clonal evolution trees: clevRact (clonal evolution in R: analyzing conserved trajectories).

- Exact SNVs/indels are reduced to the gene they affect.

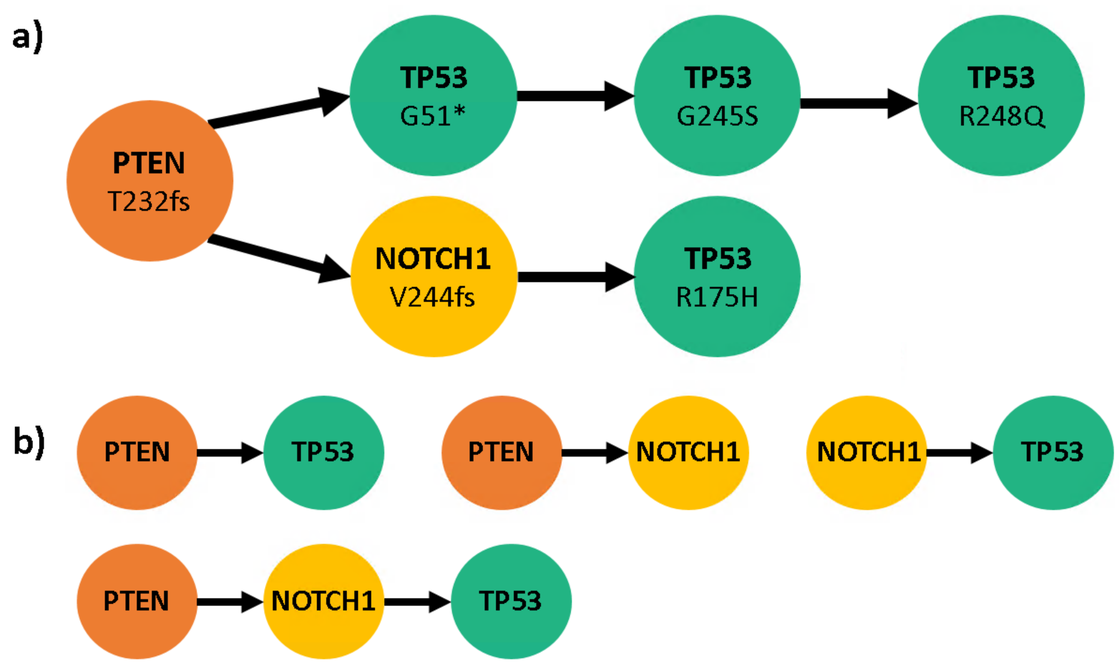

Figure 1: a) Exemplary clonal evolution. An initial mutation affecting PTEN is followed by three mutations in TP53 in one branch, and mutations in NOTCH1 and TP53 in another branch. b) clevRact determines the unique trajectories that can be derived from a given clonal evolution, focusing on the mutated genes. Trajectories of lengths 2 and 3 are considered.

- CNVs are transformed to their smallest possible representation.

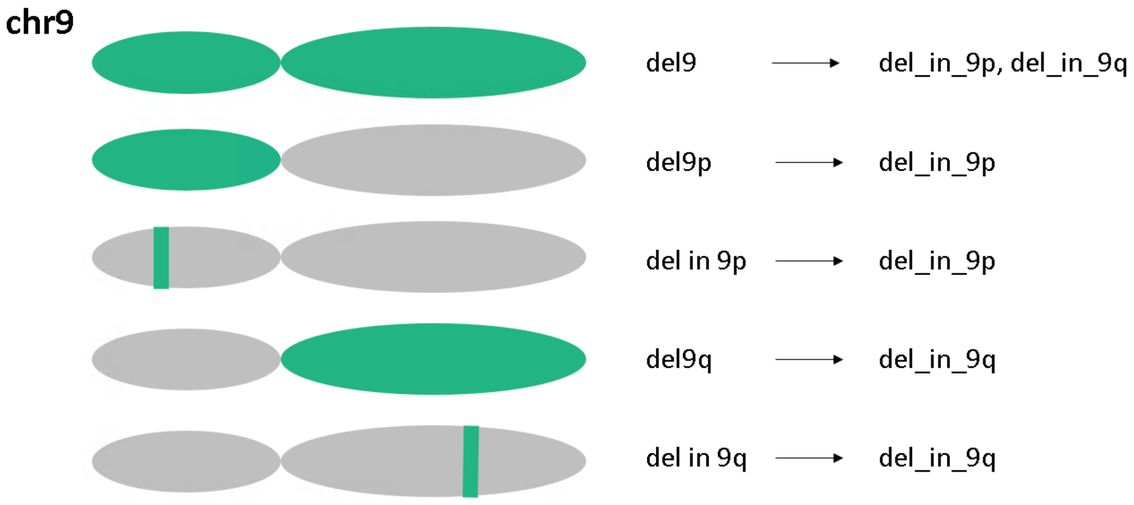

Figure 2: Exemplary CNVs affecting chromosome 9 (green). clevRact transforms del9 to its smallest possible representation, which is del_in_9p and del_in_9q. Thereby, it is possible to identify common elements of different CNVs affecting the same chromosome.

For each patient, all observed patterns of 2 and 3 levels, including both SNVs/indels and CNVs, are determined. The probability p_{ij} for observing a pattern i in patient j is calculated using Laplace-distribution.

$$p_{ij} = \frac{\#observed\ patterns}{\#possible\ patterns} \tag{1}$$

The number of observed patterns can be counted directly from the given data. The total number of possible patterns can be calculated as

$$\#possible\ patterns = \frac{m!}{(m-l)!} \tag{2}$$

with m being the number of unique mutations and l being the number of levels, i.e. either 2 or 3. Considering the example shown in Figure 1, the number of observed patterns for 2 levels is 3. The number of possible patterns is $3!/(3-2)! = 6$. Thus, the probability of observing e.g. PTEN $\rightarrow$ TP53 is $3/6 = 0.5$.

Considering a group of n patients, for every pattern i observed in this group, a vector of probabilities $P_{i*} = [p_{i1}, p_{i2}, ..., p_{in}]$ can be defined. We count c_i, the number of times a pattern i is observed across all patients n. The vector P_{i*} is assumed to follow a Poison binomial distribution, as it is the sum of n independent and not identically distributed Binomial variables. To consider statistical significance, the probability for observing pattern i in $\geq c_i$ out of n patients is calculated.

Results

We performed an exemplary analysis on a set of 149 patients with T-cell lymphoblastic neoplasms (split into 7 subgroups), including several cases of hypermutation. SNVs/indels were called in bulk DNA-sequencing data, CNVs were determined using SNP arrays. Application of TreeMHN for comparison was not possible, as 91% of the samples contained clones with > 1 mutation.

Performing analysis with MASTRO, different configurations of the input were investigated. When ignoring CNVs and including several mutations affecting the same gene as e.g. $NOTCH1_a$, $NOTCH1_b$, the analysis failed due to insufficient storage requirements even for the smallest subgroup of $n = 7$. When adding CNV calls in their smallest possible representation to the input as if they were genes, e.g. del_in_9, analysis failed for 2 out of 7 subgroups ($n = 31$ and $n = 35$). We performed detailed analysis of the successfully analyzed subgroups. Substantial differences between the number of times a pattern is observed vs the number of times it is included in the traj_occ_list, which is decisive for p-value calculation, were observed, but could not be retraced.

Using clevRact, significantly enriched mutational patterns could be successfully identified for all subsets. Manual comparison to the original data proved validity of the results.

Discussion

clevRact provides an approach for identifying conserved trajectories in clonal evolution trees. By combining information on SNVs/indels and CNVs, a thorough analysis of a tumor's mutational landscape can be performed. Future work on clevRact will involve

- modeling multi-hit events,
- exploring options for exact mutation analysis,
- additionally including targeted genes overlapping CNVs.

References

[1] Luo XG *et al.* Joint inference of exclusivity patterns and recurrent trajectories from tumor mutation trees. *Nat Commun.* 2023;14(1):3676. **[2]** Pellegrina L and Vandin F. Discovering significant evolutionary trajectories in cancer phylogenies. *Bioinformatics.* 2022;38(Suppl_2):ii49-ii55.

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